

lean4ml

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Theorem 1 (Smale 1958). *There is a homotopy of immersions of S^2 into $\mathbb{R}^3$ from the inclusion map to the antipodal map $a : q \mapsto -q$.*

Proof. This obviously follows from what we did so far. □

TaylorAndSmoothness

Throughout this section, E is a real Hilbert space, $f : E \rightarrow \mathbb{R}$, $s \subseteq E$, and $L \in \mathbb{R}_{\geq 0}$ (Lean: `NNReal`).

Definition 2 (L -smoothness on a set). We say that f is L -smooth on s if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|, \quad \forall x, y \in s.$$

In Lean: `LSmoothOn f L s := LipschitzOnWith L (fun x => gradient f x) s`.

Definition 3 (Global L -smoothness). A function is globally L -smooth if ∇f is L -Lipschitz on all of E . In Lean: `LSmoothfn f L := LipschitzWith L (fun x => gradient f x)`.